Ideation Phase
Brainstorm & Idea Prioritization Template

Date	26 June 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Reference: https://www.mural.co/templates/brainstorm-and-idea-prioritization

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Team Gathering & Collaboration:

The team came together to review the day-to-day challenges farmers, agronomists, and policymakers face when deciding which crop to grow under a given set of soil and climate conditions. Through discussion, the team identified a clear opportunity to apply data-driven decision support to agriculture — using soil nutrient levels, weather parameters, and crop history to guide better farming decisions.

Problem Statement:

Farmers often lack access to a reliable, data-driven tool that recommends the most suitable crop based on soil nutrient levels (Nitrogen, Phosphorous, Potassium), pH, temperature, humidity, and rainfall. This leads to suboptimal crop selection, reduced yields, inefficient use of resources, and limited support for researchers and policymakers studying crop-environment relationships. There is a need for an intelligent system — **OptiCrop** — that analyzes environmental and soil parameters to recommend optimal crops, assess crop-environment suitability, and support data-driven agricultural research and policy planning.

Step-2: Brainstorm, Idea Listing and Grouping

Idea Group	Brainstormed Ideas
Data Input & Preprocessing	Collect N, P, K, pH, temperature, humidity & rainfall inputs; validate and clean manual/sensor entries; support CSV and IoT sensor integration; handle missing or outlier values.

Idea Group	Brainstormed Ideas
Crop Recommendation Engine	ML classification model (e.g. Random Forest, XGBoost, Decision Tree) trained on crop data; rank top-N suitable crops; confidence score per recommendation; explainable feature-importance insights.
Suitability & Environmental Assessment	Compare a chosen crop against ideal parameter ranges; suitability score/percentage; visual indicators (gauge or radar chart); suggested corrective actions (e.g. fertilizer adjustment).
Research & Policy Analytics	Dashboards for regional crop-environment trend analysis; correlation analysis between soil parameters and yield; exportable reports for policymakers; historical trend visualization.
User Experience & Accessibility	Simple web/mobile entry form for farmers; multilingual support; voice or SMS-based input for low-literacy users; offline-first design for rural connectivity.

Step-3: Idea Prioritization

Idea	Impact	Feasibility	Priority
ML-based crop recommendation engine	High	High	P1
Data validation & preprocessing pipeline	High	High	P1
Suitability / environmental assessment scoring	High	Medium	P1
Research / policy analytics dashboard	Medium	Medium	P2
Explainable AI insights (feature importance)	Low	Medium	P3
Multilingual / voice / SMS input support	Medium	Low	P3
IoT sensor integration	Medium	Low	P3